Beyond the Single Prototype: MoE, Test-Time Adaptation, Edge Purification and PPR Residuals for AnomalyGFM

Second study — what works, what breaks, and why

Term project team

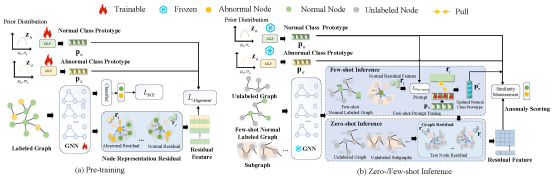
November 11, 2026

Outline

1. Context
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4. C2: TTA
5. C3: Edge purification
6. C4: PPR
7. Process
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AnomalyGFM in one slide

- Pre-train *once* on one graph (Facebook), detect anomalies in unseen graphs zero-shot.
- Node **residual** $r_v = h_v - \sum_u P_{vu} h_u$ (1-hop neighbour mean): small for normal, large for anomalous nodes.
- **One** learnable abnormal prototype p_a; score $s_v = \exp\cos(r_v, p_a)$.
- Adjacency is trusted blindly; no adaptation to the target without labels.



Four weaknesses → four extensions

C1 MoE prototypes
K abnormal experts
+ gate + load balance

one prototype can't
cover diverse anomalies

C2 Test-time adapt.
entropy + confident
pseudo-labels, prompt only

few-shot needs labels

C3 Edge purification
Gumbel-sigmoid edge
dropping before message passing

camouflage edges
hide anomalies

C4 A
multi-
per-noc

1-
deep

Protocol (real graphs only)

- **Train:** Facebook (1,081 nodes; 30% labelled; 301 epochs; the paper's recipe)
- **Zero-shot targets:** Amazon (11,944), Amazon-UPU (10,224), Reddit (10,984), Elliptic (46,564)
- **5 seeds**, AUROC/AUPRC on all target nodes; params; peak GPU memory
- **Controls** isolate each idea (attack-augmentation without purifier; TTA with shuffled pseudo-labels)
- Attack benchmark perturbs only the *edge set* of real graphs

Infrastructure

RTX 4500 Ada (24 GB); code local, jobs run on the server (`sync.sh`); sparse GPU operators.
Baseline reproduces the first study exactly (seed 0 Amazon AUROC 0.8603).

Headline: no extension improves average AUROC

Method	Amazon	UPU	Reddit	Elliptic	**Avg AUROC**	Avg AUPRC	GPU MB
Baseline AnomalyGFM	.834	.792	.573	.394	**.648**	.244	626
+ C1 MoE (K=4)	.820	.793	.556	.399	.642	.226	630
+ C2 TTA (30 steps)	.812	.796	.551	.418	.644	.211	860
ctrl: shuffled pseudo-labels	.840	.795	.556	.417	.652	.247	860
+ C3 Edge purification	.824	.758	.500	**.499**	.645	.252	1,675
ctrl: attack-aug, no purifier	.799	.771	.528	.372	.617	.213	640
+ C4 Adaptive PPR	.834	.795	.558	.395	.646	.251	675
C1+C3+C4 combined	.825	.753	.482	.499	.640	.220	1,706

- Differences of the average are within seed noise ($\sigma = 0.009$–0.037). Params: +3,204 (C1), +80 updated (C2), +705 (C3), +132 (C4).
- What *is* robust is narrower: C3 under attack, C4's learned hop gate, and why C1/C2 fail (next slides).

C1: Mixture-of-Experts abnormal prototypes

Experts: $p_k = \mathrm{ReLU}(W_{abn}\epsilon_k + \delta_k),\ k \le K$
Gate: $g_v = \mathrm{softmax}(\mathrm{Linear}(r_v/\|r_v\|))$
Score: $s_v = \sum_k g_{vk} \cos(r_v, p_k)$

$$\mathcal{L} = \sum_k g_{vk}\|p_k - r_v\| + \lambda_{lb} K \sum_k f_k P_k + \lambda_{div} \overline{\cos(p_k, p_l)}$$

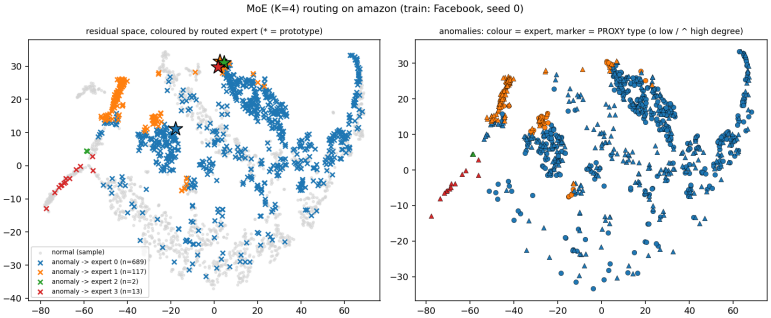
load balancing (Switch-style) + expert diversity; +3,204 params.

K	Amazon	UPU	Reddit	Ellip.
1 (paper)	**.834**	.792	**.573**	.394
2	.827	.796	.563	.405
4	.820	.793	.556	.399
8	.812	.791	.547	.406

AUROC; λ's and lr chosen on Facebook only.

- AUROC *decreases* slowly with K on Amazon/Reddit; flat elsewhere.

C1: routing collapses — t-SNE of residuals (Amazon)



Expert usage (821 anomalies):
Amazon [689, 117, 2, 13]
UPU [438, 220, 7, 28]
Reddit [366, 0, 0, 0]

Silhouette 0.145 / 0.076.

Experts 1 & 3 occupy separate regions.
Expert 0 takes 84% (Reddit: 100%).

No dataset has anomaly-type labels $\Rightarrow$ we cannot claim experts = types; assignment correlates with a degree *proxy* ($p < 10^{-4}$). Only $\approx$7 labelled anomalies in the training split.

C2: unsupervised test-time adaptation

Update only an 80-parameter input prompt on the unlabeled target:

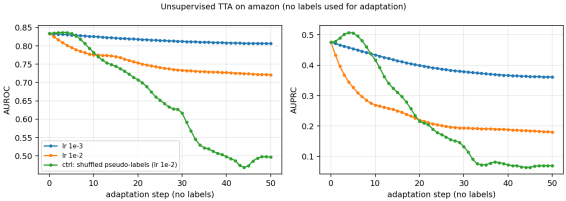
1. score $\rightarrow$ robust $z \rightarrow p_v = \sigma\big((z_v - q)/T\big)$, $q =$ assumed 5% contamination quantile
2. pseudo-labels: $p > 0.95$ anomaly, $p < 0.05$ normal
3. loss = BCE(confident) $+ 0.1 \cdot$entropy(all)

Ground truth used *only* to log AUROC/AUPRC per step.
Control: shuffled pseudo-labels.

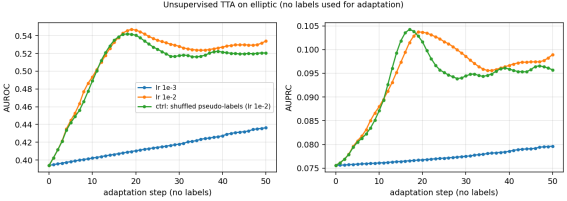
ΔAUROC	Amz	UPU	Red	E
lr 10^{-3}	$-.028$	$+.005$	$-.030$	$+.$
lr 10^{-2}	$-.113$	$-.016$	$-.085$	$+.$
shuffled ctrl	$-.338$	$-.230$	$-.055$	$+.$
pseudo-pos. prec.	.75	.65	.007	.0
base rate	.069	.068	.033	.0

C2: learning curves (no labels used for adaptation)

Amazon — pseudo-labels good (75%)



Elliptic — pseudo-labels worse than chance



- Good pseudo-labels *still* hurt (entropy sharpening). Elliptic's gain is matched by **random** pseudo-labels ⇒ not label-free learning.

C3: adversarial edge purification

Scorer (symmetric edge features $\Rightarrow w_{uv} = w_{vu}$): $|x_u - x_v|$, $x_u \odot x_v$, degrees, common neighbours + Salton overlap $\rightarrow$ MLP.

Train: Gumbel-sigmoid $w = \sigma((\ell + \eta)/\tau)$; **test:** hard drop $\ell \leq 0$.

Adversarial training: each epoch (50%) inject 10% random + 10 camouflage edges per anomaly; injected edges are known $\Rightarrow$ edge-level BCE.

Weights re-weight both the GCN operator and the neighbour mean (before the residual); 705 params.

Design history (source graph only)

	edge-AUROC	real edges d
paper lr 10^{-4}	0.30	0% (no-
node features only	0.70	48%
+ common neighbours	**0.96**	3–4%

Held-out Facebook AUROC, clean / attacked:
purified 0.839 / 0.735 vs no purifier 0.741 / 0.610.

C3: the baseline crashes under camouflage — the purifier holds

Amazon-UPU, AUROC (5 seeds)

	clean	camo 5	camo 20	camo 50	rewire 30%	rewire 50%
Baseline	.792	.669	.244	**.085**	**.822**	**.819**
Baseline + attack-aug (ctrl)	.771	.632	.164	.046	.799	.793
Edge-purified	.760	**.734**	**.701**	**.678**	.667	.617

- Camouflage (m random normal neighbours per anomaly) inverts the baseline's ranking; edge-level diagnostic: purifier AUROC **0.99**, drops 99.5% of injected vs 4.7% of real edges.
- But: worse than baseline under random rewiring (it deletes real edges too); Amazon (avg degree 736) is barely affected by either attack.

C3: where it fails — and where it helps

Reddit: fails

- Reddit has **zero common neighbours** on every edge (triangle-free).
- Facebook-learned "no shared neighbours = fake" deletes 96% of real edges; clean AUROC 0.50.
- Under camouflage 50: 0.300 vs baseline 0.029 — still < 0.5.

Elliptic: helps

- AUROC **0.499** vs baseline 0.394 and control 0.372 ($+0.105$, $+0.127$).
- Still only chance level — it removes an *inversion*, not a solution.

Cost: peak GPU 1.7 GB vs 0.6 GB (common-neighbour computation).

C4: adaptive multi-hop (PPR) residual

$$r_v = h_v - \sum_{k=1}^{K} w_{vk} \, (P^k h)_v$$

- `hop1`: paper `hop2`: 0.5/0.5
- `ppr`: $w_k \propto (1-\alpha)^{k-1}$, $\alpha = 0.3$
- `adaptive`:
  $w_v = \mathrm{softmax}\big(\log \text{prior} + \mathrm{MLP}(\text{structure}_v)\big)$,
  zero-init $\Rightarrow$ starts at PPR

K=4 sparse mat-muls, no $N \times N$ matrix; +132 params.

AUROC	Amz	UPU	Red	Ell	Tol	**A**
1-hop (paper)	.834	.792	.573	.394	.538	**.6**
2-hop	.819	.792	.531	.380	.544	.6
PPR	.815	.790	.534	.372	.540	.6
Adaptive	.834	.795	.558	.395	.532	.6

AUPRC avg: .249 / .233 / .225 / **.254**

C4: the gate learns an adaptive receptive field — but it only ties 1-hop

Mean learned hop weights	hop 1	hop 2	hop 3	hop 4
PPR prior ($\alpha = 0.3$)	.395	.276	.193	.135
Facebook (train)	.785	.003	.003	.210
Amazon	.801	.006	.005	.188
Reddit	.788	.006	.006	.200
Elliptic	.853	.002	.003	.142

- Within-graph std $\approx$ 0.27–0.30: **bimodal per node** — most nodes compare with their 1-hop mean, a minority with a 4-hop reference; same pattern on every graph.
- Fixed multi-hop references are worse (dilute the contrast). Adaptive *recovers* 1-hop accuracy but does not beat it — not even on Elliptic (laundering structures): 0.395 vs 0.394.
- P^k is a walk operator: even hops include returns to the node.

A process lesson: round 1 vs round 2

Round 1 (new modules trained at the paper's lr 10^{-4}): everything collapsed. Diagnostics showed:

- purifier **dropped 0% of edges**; edge-AUROC 0.30 *on the training graph*
- adaptive gate $\equiv$ fixed PPR (never moved)
- controls: the damage came from the augmentation, not the purifier

Fix (selected on Facebook only): per-module learning rates, common-neighbour edge features, un-balanced edge loss, augmentation strength.

Honesty note: I saw round-1 target results before redesigning the purifier; hyper-parameters came from the source only, but round 2 is not fully blind.

Avg AUROC	Round 1	Round 2
Baseline	.648	.648
C1 MoE	.639	.642
C3 purifier	.559	.645
C4 adaptive	.626	.646
combined	.494	.640

What we can and cannot claim

Can claim

- The paper's residual is **fragile to camouflage** (ranking inverts: 0.792→0.085).
- C3 with structural edge features detects injected edges (0.99) and keeps 0.68 under 50-edge camouflage; improves Elliptic over baseline *and* control.
- An adaptive PPR gate learns per-node hop choice.
- MoE routing collapses with ≈7 labelled anomalies.
- Self-training only works when the zero-shot ranking is already good.

Cannot claim

- Any average accuracy gain.
- That MoE experts are anomaly *types* (no type labels exist).
- That C3 helps without triangles or under random noise.
- Generality beyond 4 targets / one source graph.
- A fully blind round 2.

Next steps and reproducibility

Next steps

- Graph-agnostic purifier (per-graph normalised structure; learned embeddings) — fixes Reddit
- Multi-source pre-training for MoE (more labelled anomalies)
- Consensus pseudo-labels + stopping rule for TTA
- Non-backtracking walks for C4

Reproduce (`Anomaly2/`):

`python main.py all`
(about 15 min, one RTX 4500 Ada)

`README.md`: hooks into the official code
`WORK.md`: full story
`reports/C1-C4`: details

github.com/SumitKumar-17/AnomalyGFM
(branch `team-contributions`, folder `Anomaly2/`)